

AI / ML Engineer – Complete Syllabus Roadmap

♦ PART 1: AI/ML FOUNDATION COURSE (2 Months – 8 Weeks)

Week 1 – Mathematical Foundations

- Linear Algebra: vectors, matrices, dot product, eigen concepts
 - Probability basics: events, distributions
 - Statistics: mean, variance, standard deviation
 - Calculus basics: derivatives, gradients (intuition only)
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Week 2 – Python for Data Science

- Python fundamentals (loops, functions, OOP basics)
 - NumPy (arrays, broadcasting, vectorization)
 - Pandas (dataframes, indexing, merging)
 - Jupyter Notebook workflow
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Week 3 – Data Handling & EDA

- Data cleaning (missing values, duplicates)
- Data preprocessing & scaling
- Feature engineering basics

- Exploratory Data Analysis (EDA)
 - Data visualization (Matplotlib, Seaborn)
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Week 4 – Supervised Learning (Regression)

- Machine Learning pipeline
 - Linear Regression
 - Multiple Linear Regression
 - Logistic Regression
 - Bias–variance tradeoff
 - Hands-on mini project
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Week 5 – Supervised Learning (Classification)

- Decision Trees
 - Random Forest
 - K-Nearest Neighbors (KNN)
 - Support Vector Machines (SVM)
 - Evaluation metrics (Accuracy, Precision, Recall, F1)
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Week 6 – Unsupervised Learning

- K-Means clustering
- Hierarchical clustering

- Dimensionality reduction (PCA)
 - Use cases & visualization
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Week 7 – Deep Learning Basics

- Neural Network fundamentals
 - Activation functions
 - Backpropagation intuition
 - CNN basics (image understanding)
 - RNN basics (sequence data)
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Week 8 – Model Evaluation & Deployment

- Cross-validation
 - Hyper-parameter tuning
 - Regularization techniques
 - Intro to deployment
 - Deploy ML model using **Streamlit**
 - Foundation final project
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♦ PART 2: ADVANCED AI/ML ENGINEER COURSE (5 Months – 20 Weeks)

Month 1: Advanced Math & ML (Weeks 9–12)

Week 9 – Advanced Mathematics

- Multivariate calculus
 - Optimization techniques (GD, SGD)
 - Convex optimization
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Week 10 – Bayesian & Probabilistic ML

- Bayesian inference
 - Probability models
 - Probabilistic graphical models
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Week 11 – Advanced Machine Learning

- Regularization (L1, L2, ElasticNet)
 - Kernel methods
 - Model selection strategies
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Week 12 – Ensemble Learning

- Gradient Boosting
 - XGBoost
 - LightGBM
 - Feature importance & tuning
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Month 2: Deep Learning Specialization (Weeks 13–16)

Week 13 – CNN Deep Dive

- CNN architectures
 - Image classification
 - Object detection basics
 - Transfer learning
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Week 14 – Advanced Computer Vision

- Object detection (YOLO, SSD, Faster R-CNN)
 - Image segmentation (U-Net, Mask R-CNN)
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Week 15 – RNNs & Transformers

- RNN, LSTM, GRU
 - Seq2Seq models
 - Attention mechanism
 - Transformers overview
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Week 16 – Generative Models

- GANs
- VAEs

- Diffusion models
 - Image & text generation projects
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Month 3: NLP & LLMs (Weeks 17–20)

Week 17 – NLP Fundamentals

- Text preprocessing
 - Tokenization
 - Word embeddings (Word2Vec, GloVe, FastText)
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Week 18 – Advanced NLP Tasks

- Sentiment analysis
 - Named Entity Recognition (NER)
 - Text summarization
 - Question Answering
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Week 19 – Large Language Models

- Transformer architecture deep dive
- BERT, GPT overview
- Prompt engineering
- Fine-tuning LLMs

Week 20 – LLM Applications

- Chatbots
- RAG systems
- LLM evaluation
- LLM deployment basics

Month 4: MLOps & Production (Weeks 21–24)

Week 21 – MLOps Fundamentals

- ML lifecycle
- Model versioning (DVC, MLflow)
- Experiment tracking

Week 22 – CI/CD for ML

- Git & GitHub
- CI/CD pipelines
- GitHub Actions / Jenkins for ML

Week 23 – Deployment & Scaling

- Docker

- Kubernetes basics
 - REST APIs for ML
 - Scaling inference
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Week 24 – Monitoring & Reliability

- Model monitoring
 - Data & concept drift
 - Logging & alerting
 - Production best practices
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Month 5: Special Topics & Capstone (Weeks 25–28)

Week 25 – Advanced Topics

- Graph Neural Networks (GNNs)
 - Time-Series forecasting
 - Anomaly detection
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Week 26 – Ethical & Responsible AI

- Bias & fairness
- Explainable AI
- Privacy-preserving ML

- Federated learning & DP
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Week 27 – Capstone Project (Build Phase)

- Problem definition
 - Data pipeline
 - Model development
 - Evaluation
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Week 28 – Capstone Project (Deploy & Present)

- End-to-end deployment
- Documentation
- Presentation & demo